



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 9

1. Case Study Title: Common Case Study Scenario for Practical 1-8

Develop a Student Task Management Web Application and apply Dark Theme and Light Theme functionality across all Practical 1-8.

The application should allow users to perform the activities covered in Practical 1-8, such as creating and managing tasks, updating content using DOM operations, handling user interactions, validating inputs, and dynamically modifying webpage elements.

- A Theme Toggle Button should be included in every practical application, allowing users to switch between Light Mode and Dark Mode. The selected theme should be saved using LocalStorage, so the user's preference remains available even after closing and reopening the browser.

2. Case Study Program Code:

```
<!DOCTYPE html>
<!-- Sonal Nakhate 24070521199 -->
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Form</title>
```



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```
<style>
  /* CSS Theme Variables */
  :root {
    --bg-color: #f4f4f9;
    --text-color: #333333;
    --card-bg: #ffffff;
    --border-color: #cccccc;
    --input-bg: #ffffff;
    --label-color: #555555;
    --box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
  }

  [data-theme="dark"] {
    --bg-color: #121212;
    --text-color: #f0f0f0;
    --card-bg: #1e1e1e;
    --border-color: #444444;
    --input-bg: #2d2d2d;
    --label-color: #bbbbbb;
    --box-shadow: 0 4px 10px rgba(0, 0, 0, 0.5);
  }

  [data-theme="light"] {
    --bg-color: #f4f4f9;
    --text-color: #333333;
    --card-bg: #ffffff;
    --border-color: #cccccc;
    --input-bg: #ffffff;
    --label-color: #555555;
    --box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
  }

  body {
    font-family: Arial, sans-serif;
```



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```
background-color: var(--bg-color);
color: var(--text-color);
display: flex;
flex-direction: column;
justify-content: center;
align-items: center;
min-height: 100vh;
margin: 0;
padding: 20px 0;
box-sizing: border-box;
transition: background-color 0.3s, color 0.3s;
}

.theme-selector {
  width: 100%;
  max-width: 400px;
  margin-bottom: 15px;
}

.form-container {
  background: var(--card-bg);
  padding: 25px 30px;
  border-radius: 8px;
  box-shadow: var(--box-shadow);
  width: 100%;
  max-width: 400px;
  border: 1px solid var(--border-color);
  box-sizing: border-box;
  transition: background-color 0.3s, border-color 0.3s;
}

h2 {
  text-align: center;
  color: var(--text-color);
}
```



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```
margin-bottom: 20px;
}

.form-group {
  margin-bottom: 15px;
}

label {
  display: block;
  font-weight: bold;
  margin-bottom: 5px;
  color: var(--label-color);
}

input, select {
  width: 100%;
  padding: 10px;
  border: 1px solid var(--border-color);
  border-radius: 4px;
  box-sizing: border-box;
  font-size: 14px;
  background-color: var(--input-bg);
  color: var(--text-color);
}

input:focus, select:focus {
  outline: none;
  border-color: #007bff;
}

.error {
  color: #dc3545;
  font-size: 12px;
  margin-top: 5px;
}
```



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```
display: block;
min-height: 15px;
}

.valid {
  border-color: #28a745 !important;
}

.invalid {
  border-color: #dc3545 !important;
}

button {
  width: 100%;
  padding: 12px;
  background-color: #007bff;
  color: white;
  border: none;
  border-radius: 4px;
  font-size: 16px;
  font-weight: bold;
  cursor: pointer;
}

button:hover {
  background-color: #0056b3;
}

.success-msg {
  color: #28a745;
  text-align: center;
  font-weight: bold;
  margin-top: 15px;
}
```



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```
</style>
</head>
<body>

<!-- Theme Selector Control -->
<div class="theme-selector">
  <label for="themeSelect"><strong>Theme: </strong></label>
  <select id="themeSelect" onchange="applyTheme(this.value)">
    <option value="default" selected>System Default</option>
    <option value="light">Light</option>
    <option value="dark">Dark</option>
  </select>
</div>

<div class="form-container">
  <h2>Gym Admission Form</h2>
  <form id="gymForm" novalidate>

    <!-- Full Name -->
    <div class="form-group">
      <label for="fullName">Full Name</label>
      <input type="text" id="fullName" placeholder="Enter your full name">
      <span class="error" id="nameError"></span>
    </div>

    <!-- Email Address -->
    <div class="form-group">
      <label for="email">Email Address</label>
      <input type="email" id="email" placeholder="abc@xyz.com">
      <span class="error" id="emailError"></span>
    </div>

    <!-- Mobile Number -->
    <div class="form-group">
```



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```
<label for="mobile">Mobile Number</label>
<input type="tel" id="mobile" placeholder="10-digit mobile number">
<span class="error" id="mobileError"></span>
</div>

<!-- Age -->
<div class="form-group">
  <label for="age">Age (18 - 50)</label>
  <input type="number" id="age" placeholder="Enter your age">
  <span class="error" id="ageError"></span>
</div>

<!-- Membership Plan -->
<div class="form-group">
  <label for="plan">Membership Plan</label>
  <select id="plan">
    <option value="">-- Select a Plan --</option>
    <option value="quarterly">Quarterly</option>
    <option value="half-yearly">Half Yearly</option>
    <option value="annually">Annually</option>
  </select>
  <span class="error" id="planError"></span>
</div>

<button type="submit">Submit Admission</button>
<div id="successMsg" class="success-msg"></div>
</form>
</div>

<script>
  // Theme Switcher Logic
  function applyTheme(theme) {
    const root = document.documentElement;
    if (theme === "default") {
```



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```
const prefersDark = window.matchMedia("(prefers-color-scheme:
dark)").matches;

root.setAttribute("data-theme", prefersDark ? "dark" : "light");
} else {
root.setAttribute("data-theme", theme);
}
}

// Apply default theme on page load
applyTheme("default");

// Get element references
const fullName = document.getElementById('fullName');
const email = document.getElementById('email');
const mobile = document.getElementById('mobile');
const age = document.getElementById('age');
const plan = document.getElementById('plan');
const gymForm = document.getElementById('gymForm');

// Helper function to update input styling and error message
function setValidationState(input, errorElement, isValid, errorMessage)
{
    if (isValid) {
        input.classList.remove('invalid');
        input.classList.add('valid');
        errorElement.textContent = '';
        return true;
    } else {
        input.classList.remove('valid');
        input.classList.add('invalid');
        errorElement.textContent = errorMessage;
        return false;
    }
}
```



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```
// --- Live Validation Functions ---

function validateName() {
  const value = fullName.value.trim();
  const isValid = value.length >= 3;
  return setValidationState(
    fullName,
    document.getElementById('nameError'),
    isValid,
    'Name must be at least 3 characters long.'
  );
}

function validateEmail() {
  const value = email.value.trim();
  // Basic email pattern regex
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  const isValid = emailRegex.test(value);
  return setValidationState(
    email,
    document.getElementById('emailError'),
    isValid,
    'Please enter a valid email address.'
  );
}

function validateMobile() {
  const value = mobile.value.trim();
  // Validates a 10-digit phone number
  const mobileRegex = /^[0-9]{10}$/;
  const isValid = mobileRegex.test(value);
  return setValidationState(
    mobile,
```



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```
document.getElementById('mobileError'),
isValid,
'Please enter a valid 10-digit mobile number.'
);
}

function validateAge() {
  const value = parseInt(age.value, 10);
  const isValid = !isNaN(value) && value >= 18 && value <= 50;
  return setValidationState(
    age,
    document.getElementById('ageError'),
    isValid,
    'Age must be between 18 and 50.'
  );
}

function validatePlan() {
  const isValid = plan.value !== '';
  return setValidationState(
    plan,
    document.getElementById('planError'),
    isValid,
    'Please select a membership plan.'
  );
}

// --- Attach Event Listeners for Live Checks ---

fullName.addEventListener('input', validateName);
email.addEventListener('input', validateEmail);
mobile.addEventListener('input', validateMobile);
age.addEventListener('input', validateAge);
plan.addEventListener('change', validatePlan);
```



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```
// 'submit' event to validate all fields before sending form data
gymForm.addEventListener('submit', function (event) {
    event.preventDefault(); // Prevent page reload

    // Run all validations
    const isNameValid = validateName();
    const isEmailValid = validateEmail();
    const isMobileValid = validateMobile();
    const isAgeValid = validateAge();
    const isPlanValid = validatePlan();

    // Check if all fields are valid
    if (isNameValid && isEmailValid && isMobileValid && isAgeValid &&
isPlanValid) {
        document.getElementById('successMsg').textContent = 'Admission
Submitted Successfully!';
        gymForm.reset(); // Reset fields

        // Clear visual validation indicators
        [fullName, email, mobile, age, plan].forEach(input => {
            input.classList.remove('valid');
        });
    } else {
        document.getElementById('successMsg').textContent = '';
    }
});
</script>

</body>
</html>
```



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3. Output:

The screenshot shows a web browser window with the URL `C:/Users/Sonal/Desktop/Documents/JS_24070521199/Exp9/CS8.html`. The page has a dark background. At the top, there is a 'Theme:' dropdown menu set to 'Dark'. Below it is a 'Gym Admission Form' with the following fields: 'Full Name' (placeholder: 'Enter your full name'), 'Email Address' (placeholder: 'abc@xyz.com'), 'Mobile Number' (placeholder: '10-digit mobile number'), 'Age (18 - 50)' (placeholder: 'Enter your age'), and 'Membership Plan'.

The screenshot shows the same web browser window with the same URL. The page has a light background. The 'Theme:' dropdown menu is now set to 'Light'. The 'Gym Admission Form' fields are the same as in the previous screenshot: 'Full Name' (placeholder: 'Enter your full name'), 'Email Address' (placeholder: 'abc@xyz.com'), 'Mobile Number' (placeholder: '10-digit mobile number'), 'Age (18 - 50)' (placeholder: 'Enter your age'), and 'Membership Plan'.



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4. Result/Conclusion: Dynamic real-time form validation was successfully executed using `input`, `change`, and `submit` event listeners without forcing full page reloads.

Flexible theme switching was successfully integrated using CSS variables and JS-driven `data-theme` attribute manipulation on the root HTML element.